

# National University of Computer and Emerging Sciences, Lahore Campus



|             |                                    |              |             |
|-------------|------------------------------------|--------------|-------------|
| Course:     | Parallel and Distributed Computing | Course Code: | CS-3006     |
| Program:    | BSCS & BSDS                        | Semester:    | Spring 2024 |
| Duration:   | 20 Minutes                         | Total Marks: | 14          |
| Paper Date: | 02-Apr-2024                        | Weight       | 3.34%       |
| Type:       | Announced                          | Page(s):     | 2           |
| Exam:       | Quiz 3A Solution                   | Section:     |             |

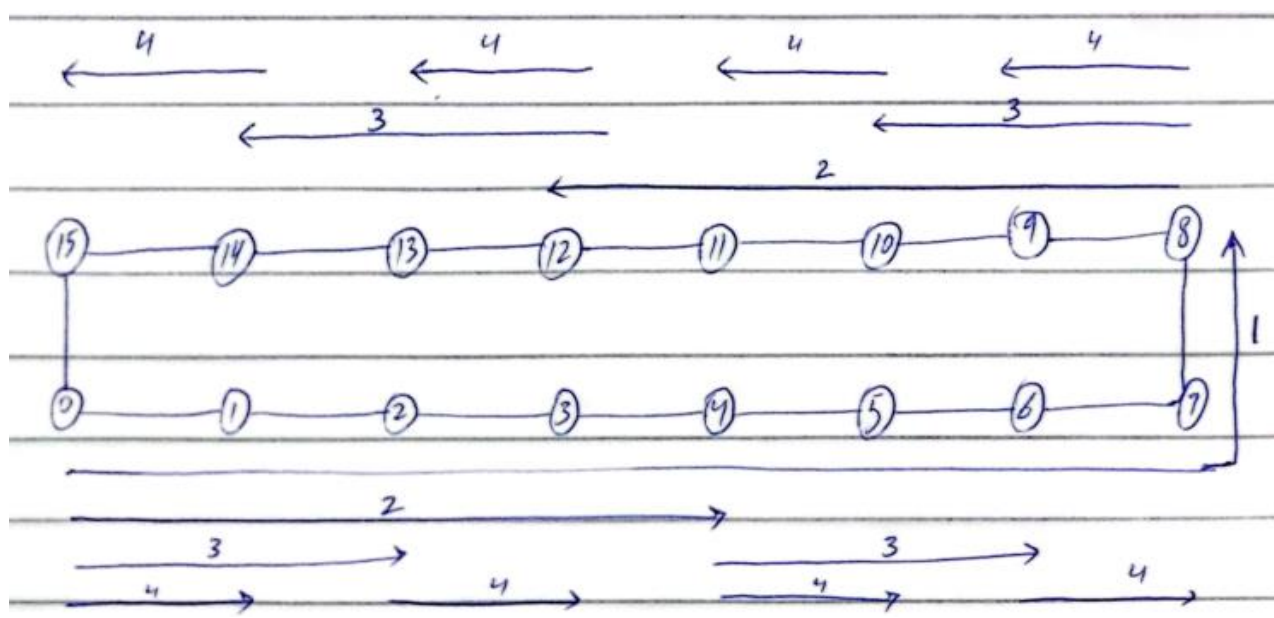
Name & Roll No: \_\_\_\_\_

Attempt all questions on the question booklet. Rough sheets can be used but it should not be attached. State answers in readable handwriting and with the help of diagrams, where necessary.

## Question # 1:

[2 + 4 marks, CLO # 3]

Draw 16 nodes ring structure. Considering initial message on node 0, perform one-to-all broadcast on 16 nodes ring structure. Show all the steps involved in the one-to-all broadcast operation.

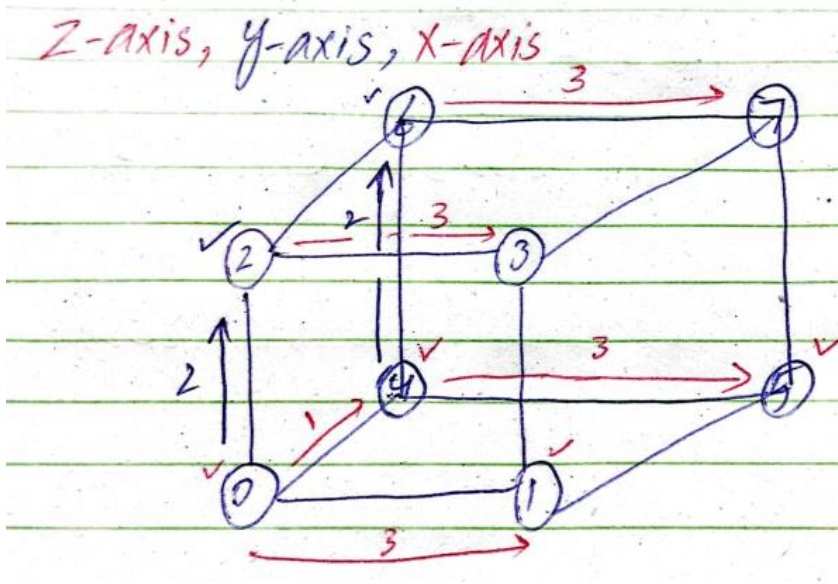


## Question # 2:

[4 marks, CLO # 3]

Draw 8 nodes hypercube structure. Considering initial message on node 0, perform one-to-all broadcast on 8 nodes hypercube structure. Show all the steps involved in the one-to-all broadcast operation.

Rule of thumb: z-axis, y-axis, x-axis. However, different order of dimensions can also be used.



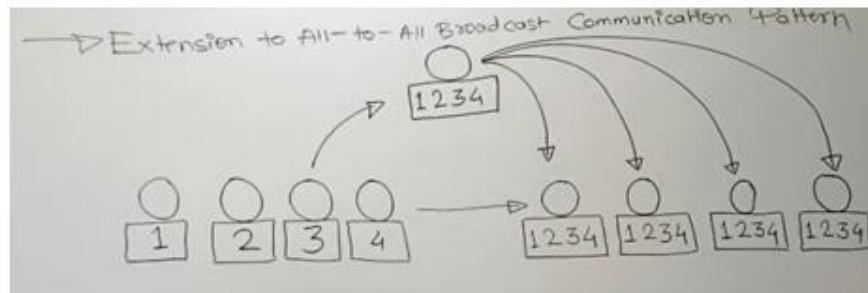
### Question # 3:

[2 + 2 marks, CLO # 3]

How is all-reduce different from total-exchange? Explain with the help of figures.

#### All-Reduce:

- Use all-to-one reduction followed by one-to-all broadcast.



#### Total Exchange:

- Each node sends a distinct message of size  $m$  to every other node.

